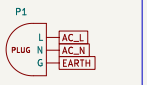
MAINS BOX

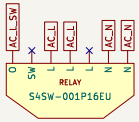
MAINS PLUG IN



NEMA 5-15P, 14/3 SJT cord

Plugs into the wall outlet.

SHELLY 1PM GEN4

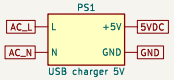


RC SNUBBER



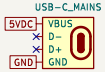
Across the load (O to N), never across the relay contacts.
C1: X2, exactly 100nF (no discharge resistor), 275VAC.
R1: must have a cement or ceramic body, not a coated film one.

ALWAYS-ON 5VDC PSU

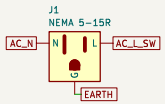


PS1 is fed before U1.

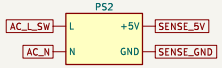
USB-C POWER OUT



NEMA 15P RECEPTACLE

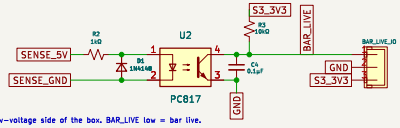


5VDC PSU FOR BAR SENSE



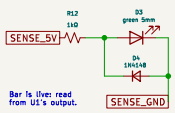
Wired in after U1 (Wagos).
Powers U2 and D3 only.

BAR SENSE



Low-voltage side of the box. BAR_LIVE low = bar live.

BAR LIVE LED



Bar is live: read from U1's output.

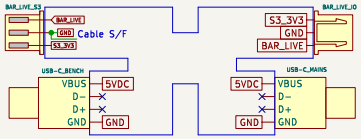
NOTES

2. Mains joints: Wago connectors. EARTH runs straight from P1 to J1: the Shelly has no earth terminal.

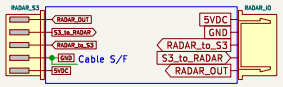
6. PS1 is fed before U1: the bench node outlives a cut.

CABLES

BENCH NODE TO MAINS BOX



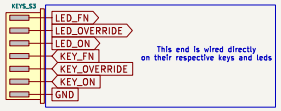
BENCH NODE TO LD2410C RADAR



BENCH NODE – S3 to BUZZER



BENCH NODE – S3 to KEYS and LEDS

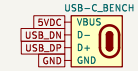


BENCH NODE – S3 to OLED



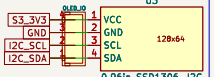
KEYPAD (BENCH NODE)

USB POWER AND DATA IN



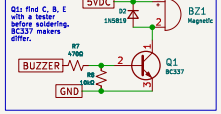
P2: short pigtail into U5's native USB port.

OLED SSD1306

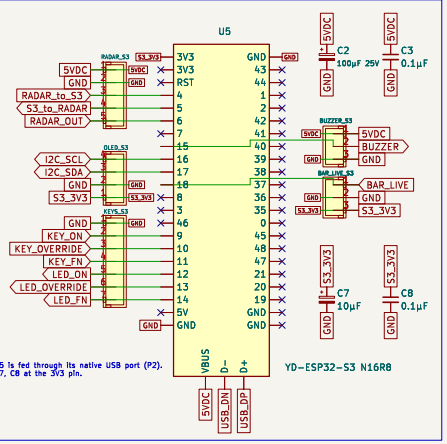


0.96in SSD1306, I2C

MAGNETIC BUZZER AND DRIVER

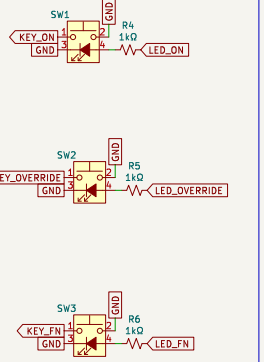


MAIN MCU ESP32-S3 N16R8



U5 is fed through its native USB port (P2).
C7, C8 at the 3V3 pin.

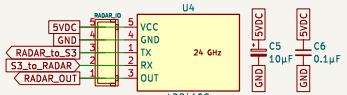
KEYS



SW1–SW3: Cherry MX with a 3mm LED. Key to GND, internal pull-up.

SENSOR NODE

RADAR LD2410C



Sheet: /
File: Power Protector.kicad_sch

Title: Power Protector

Size: A3	Date: 2026-10-25	Rev: A
KiCad E.D.A. 9.0.7		Id: 1/1